MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 08 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Questions

MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. In $\triangle ABC$, D is a point on side BC such that $\angle ADC = \angle BAC$. If $CA = 12$ cm and $CB = 16$ cm, then CD is: [1]

(a) 8 cm (b) 9 cm (c) 10 cm (d) 6 cm

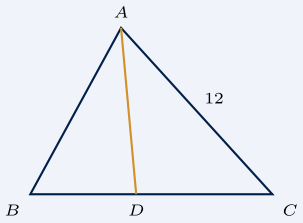
▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 9 cm

Given: $\angle ADC = \angle BAC$, $CA = 12$ cm, $CB = 16$ cm.

In $\triangle ABC$ and $\triangle DAC$: $\angle C = \angle C$ (common) and $\angle BAC = \angle ADC$ (given), so $\triangle ABC \sim \triangle DAC$ (AA similarity):

Formula / Theorem Used: $\triangle ABC \sim \triangle DAC \implies \dfrac{CB}{CA} = \dfrac{CA}{CD} \implies CA^2 = CB \cdot CD$



$$12^2 = 16 \cdot CD \implies 144 = 16 \cdot CD \implies CD = \frac{144}{16} = \boxed{9 \text{ cm}}$$

Q2. If $\triangle ABC \sim \triangle PQR$ with perimeters 35 cm and 45 cm respectively, and $PQ = 9$ cm, then AB is: [1]

(a) 7 cm (b) 8 cm (c) 6 cm (d) 7.5 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 7 cm

We know that the ratio of perimeters of two similar triangles equals the ratio of their corresponding sides:

Formula / Theorem Used: $\dfrac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle PQR)} = \dfrac{AB}{PQ}$

$$\frac{35}{45} = \frac{AB}{9} \implies \frac{7}{9} = \frac{AB}{9} \implies \boxed{AB = 7 \text{ cm}}$$

Q3. In $\triangle ABC$, $DE \parallel BC$. If $AD = 4$ cm, $AE = 8$ cm, $DB = x - 4$, and $EC = 3x - 19$, then x equals: [1]

(a) 9 (b) 10 (c) 11 (d) 12

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (c) 11

We know the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: Since $DE \parallel BC$, $\dfrac{AD}{DB} = \dfrac{AE}{EC}$

$$\frac{4}{x-4} = \frac{8}{3x-19} \implies \frac{1}{x-4} = \frac{2}{3x-19}$$

$$3x - 19 = 2(x-4) = 2x - 8 \implies x = 19 - 8 = 11$$

Verification: $DB = 11 - 4 = 7 > 0$ and $EC = 33 - 19 = 14 > 0$. Hence $\boxed{x = \mathbf{11}}$.

Q4. Two vertical poles of height 6 m and 11 m stand on level ground. If their feet are 12 m apart, distance between their tops is: [1]

(a) 12 m (b) 13 m (c) 14 m (d) 15 m

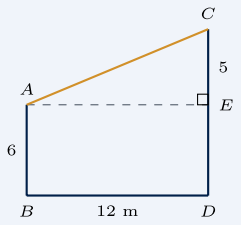
▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) 13 m

Given: Pole heights $AB = 6$ m and $CD = 11$ m, horizontal separation $BD = 12$ m. Draw $AE \perp CD$, forming right $\triangle AEC$ with $AE = 12$ m and $CE = 11 - 6 = 5$ m.

We know the Pythagoras Theorem for right-angled triangles:

Formula / Theorem Used: $AC^2 = AE^2 + CE^2$



$$AC = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = \boxed{\mathbf{13}\text{ m}}$$

Q5. In $\triangle ABC$, AD is the bisector of $\angle A$ meeting BC at D. If $AB = 6$ cm, $AC = 4$ cm, and $BD = 3$ cm, then CD is: [1]

(a) 2 cm (b) 2.5 cm (c) 3 cm (d) 1.5 cm

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) 2 cm

We know the Internal Angle Bisector Theorem (derived using BPT by drawing a line through C parallel to DA):

Formula / Theorem Used: The internal bisector of an angle of a triangle divides the opposite side in the ratio of the remaining two sides: $\dfrac{AB}{AC} = \dfrac{BD}{CD}$.

$$\frac{6}{4} = \frac{3}{CD} \implies 6 \cdot CD = 4 \times 3 = 12 \implies CD = \frac{12}{6} = \boxed{\mathbf{2}\text{ cm}}$$

Q6. **Assertion (A):** Two similar triangles are always congruent.
Reason (R): Two congruent triangles are always similar. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (d) Assertion (A) is false but Reason (R) is true.

We know the distinction between similarity and congruence:

Formula / Theorem Used: Similarity requires equal angles and proportional sides ($k > 0$); congruence requires exact equality ($k = 1$).

Two similar triangles can have different sizes (e.g., sides $3, 4, 5$ and $6, 8, 10$), so they need not be congruent $\implies$ **(A) is false**. Two congruent triangles always have equal angles and side ratio $1 : 1$, so they are always similar $\implies$ **(R) is true**.

SECTION B: Very Short Answer Questions MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. A vertical stick of length 1.8 m casts a shadow 1.2 m long on level ground. At the same time, a vertical building casts a shadow 24 m long. Find the height of the building. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Stick height $AB = 1.8$ m, stick shadow $BC = 1.2$ m; building shadow $QR = 24$ m. Let building height be $PQ = H$ m.

At the same time of day, sunlight strikes ground at equal angles ($\angle C = \angle R$), forming similar right triangles:

Formula / Theorem Used: $\triangle ABC \sim \triangle PQR$ (AA similarity) $\implies \dfrac{\text{Height of Building}}{\text{Shadow of Building}} = \dfrac{\text{Height of Stick}}{\text{Shadow of Stick}}$

Step 1: Establishing Similarity & Ratio Equation [1 Mark]

In $\triangle ABC$ and $\triangle PQR$: $\angle B = \angle Q = 90^\circ$ and $\angle C = \angle R \implies \triangle ABC \sim \triangle PQR$ (AA similarity).

$$\frac{H}{24} = \frac{1.8}{1.2}$$

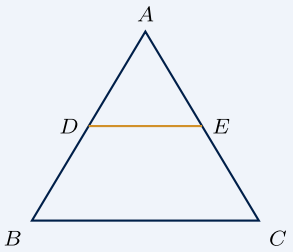
Step 2: Calculating Height of the Building [1 Mark]

Simplifying $\dfrac{1.8}{1.2} = \dfrac{18}{12} = \dfrac{3}{2}$:

$$\frac{H}{24} = \frac{3}{2} \implies H = \frac{3}{2} \times 24 = 3 \times 12 = \boxed{\mathbf{36}\ \text{m}}$$

Q8. In $\triangle ABC$, D and E are points on sides AB and AC respectively such that $\angle B = \angle C$ and $BD = CE$. Prove mathematically that $DE \parallel BC$. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]



Given: In $\triangle ABC$, D on AB, E on AC, $\angle B = \angle C$, and $BD = CE$.
To Prove: $DE \parallel BC$.
We know the isosceles triangle theorem and Converse of BPT:

Formula / Theorem Used: Sides opposite equal angles are equal; if $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$.

Step 1: Showing $AD = AE$ [1 Mark]
In $\triangle ABC$, since $\angle B = \angle C$, sides opposite equal angles give $AC = AB$, i.e., $AB = AC$.
Subtracting given equal segments $BD = CE$:

$$AB - BD = AC - CE \implies AD = AE$$

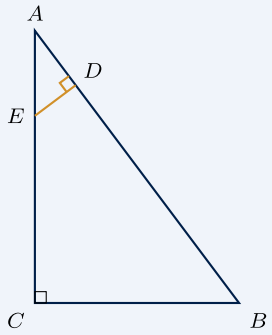
Step 2: Applying Converse of BPT [1 Mark]
Dividing equal positive lengths ($AD = AE$, $DB = EC$):

$$\frac{AD}{DB} = \frac{AE}{EC} \implies \boxed{\therefore DE \parallel BC \quad \text{(Converse of BPT)}}$$

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Q9. In the given figure, ABC is a triangle right-angled at C and $DE \perp AB$. Prove that $\triangle ABC \sim \triangle ADE$ and find the lengths of AE and DE if $AB = 15$ cm, $AC = 12$ cm, and $AD = 3$ cm. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: $\triangle ABC$ is right-angled at C ($\angle ACB = 90°$), D on AB, E on AC with $DE \perp AB$ ($\angle ADE = 90°$); $AB = 15$ cm, $AC = 12$ cm, $AD = 3$ cm.
We know Pythagoras Theorem and the AA Similarity Criterion:

Formula / Theorem Used: Exact vertex correspondence fixes equal angles: $A \leftrightarrow A$, $\angle ACB = \angle ADE = 90° \implies B \leftrightarrow E$, $C \leftrightarrow D$.

Step 1: Proving Similarity with Exact Correspondence [1 Mark]
Comparing $\triangle ABC$ and $\triangle AED$: (1) $\angle A = \angle A$ (common), and (2) $\angle ACB = \angle ADE = 90°$. By **AA similarity criterion**, $\triangle ABC \sim \triangle AED$ (referred to in the problem statement as unordered vertices $\triangle ADE$).

Step 2: Finding BC by Pythagoras Theorem [0.5 Mark]

$$BC = \sqrt{AB^2 - AC^2} = \sqrt{15^2 - 12^2} = \sqrt{225 - 144} = \sqrt{81} = 9 \text{ cm}$$

Step 3: Calculating AE and DE [1.5 Marks]
From $\triangle ABC \sim \triangle AED$, corresponding sides are proportional:

$$\frac{AB}{AE} = \frac{AC}{AD} = \frac{BC}{DE} \implies \frac{15}{AE} = \frac{12}{3} = \frac{9}{DE}$$

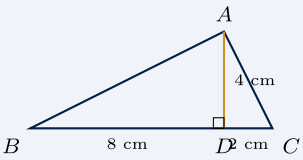
Since $\frac{12}{3} = 4$:

$$AE = \frac{15}{4} = 3.75 \text{ cm}, \qquad DE = \frac{9}{4} = 2.25 \text{ cm}$$

$$\boxed{AE = \mathbf{3.75} \text{ cm } \left(\frac{15}{4} \text{ cm}\right), \qquad DE = \mathbf{2.25} \text{ cm } \left(\frac{9}{4} \text{ cm}\right)}$$

Q10. In $\triangle ABC$, AD is an altitude to BC. If $BD = 8$ cm, $DC = 2$ cm, and $AD = 4$ cm, prove that $\triangle ABC$ is right-angled at A. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: $AD \perp BC$, $BD = 8$ cm, $DC = 2$ cm, $AD = 4$ cm.
To Prove: $\angle BAC = 90°$.
We know the SAS similarity criterion and Triangle Angle Sum Property:

> **Formula / Theorem Used:** If $\frac{BD}{AD} = \frac{AD}{DC}$ and $\angle ADB = \angle CDA = 90°$, then $\triangle ADB \sim \triangle CDA$.

Step 1: Establishing $\triangle ADB \sim \triangle CDA$ [1.5 Marks]
Comparing ratios: $\frac{BD}{AD} = \frac{8}{4} = 2$ and $\frac{AD}{DC} = \frac{4}{2} = 2 \implies \frac{BD}{AD} = \frac{AD}{DC}$.
Also $\angle ADB = \angle CDA = 90°$ (altitude $AD \perp BC$). By **SAS similarity**, $\triangle ADB \sim \triangle CDA$.

Step 2: Concluding $\angle BAC = 90°$ [1.5 Marks]
From similarity, corresponding angles are equal: $\angle BAD = \angle C$ and $\angle B = \angle CAD$.
In $\triangle ABC$, by the angle sum property:

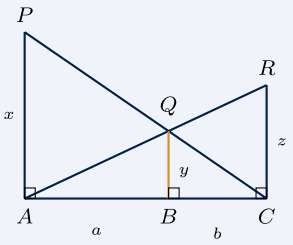
$$\angle BAC + \angle B + \angle C = 180° \implies \angle BAC + (\angle CAD + \angle BAD) = 180°$$

$$2\angle BAC = 180° \implies \boxed{\therefore \angle BAC = \mathbf{90°} \quad \text{[Hence Proved]}}$$

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Q11. In the given figure, PA, QB, RC are each perpendicular to line segment AC. If $PA = x$, $QB = y$, $RC = z$, $AB = a$, $BC = b$, prove that: (i) $\frac{y}{x} = \frac{b}{a+b}$, (ii) $\frac{y}{z} = \frac{a}{a+b}$, (iii) $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Given: $PA \perp AC$, $QB \perp AC$, $RC \perp AC$; $PA = x$, $QB = y$, $RC = z$, $AB = a$, $BC = b$, so total base $AC = AB + BC = a + b$.
We know the AA Similarity Criterion:

Formula / Theorem Used: Two right-angled triangles sharing an acute angle are similar, making corresponding sides proportional.

Step 1: Proving (i) $\frac{y}{x} = \frac{b}{a+b}$ [2 Marks]

In $\triangle CBQ$ and $\triangle CAP$: $\angle CBQ = \angle CAP = 90°$ (given $QB, PA \perp AC$) and $\angle BCQ = \angle ACP$ (common angle $\angle C$).
By **AA similarity**, $\triangle CBQ \sim \triangle CAP$. Thus:

$$\frac{QB}{PA} = \frac{BC}{AC} \implies \boxed{\frac{y}{x} = \frac{b}{a+b}} \quad \text{— (i)}$$

Step 2: Proving (ii) $\frac{y}{z} = \frac{a}{a+b}$ [2 Marks]

In $\triangle ABQ$ and $\triangle ACR$: $\angle ABQ = \angle ACR = 90°$ (given $QB, RC \perp AC$) and $\angle BAQ = \angle CAR$ (common angle $\angle A$).
By **AA similarity**, $\triangle ABQ \sim \triangle ACR$. Thus:

$$\frac{QB}{RC} = \frac{AB}{AC} \implies \boxed{\frac{y}{z} = \frac{a}{a+b}} \quad \text{— (ii)}$$

Step 3: Proving (iii) $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$ [1 Mark]

Adding equations (i) and (ii):

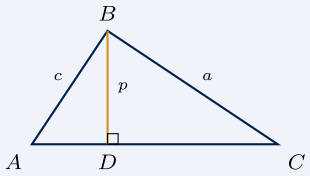
$$\frac{y}{x} + \frac{y}{z} = \frac{b}{a+b} + \frac{a}{a+b} = \frac{a+b}{a+b} = 1$$

Dividing every term by y ($y > 0$):

$$\boxed{\therefore \frac{1}{x} + \frac{1}{z} = \frac{1}{y} \quad \text{[Hence Proved]}}$$

Q12. In a right triangle ABC right-angled at B, $BD \perp AC$. Prove that $\triangle ADB \sim \triangle ABC \sim \triangle BDC$. Hence deduce that: (i) $BD^2 = AD \cdot CD$, (ii) $AB^2 = AD \cdot AC$, and (iii) if $BD = p$, $AB = c$, $BC = a$, prove that $\dfrac{1}{p^2} = \dfrac{1}{a^2} + \dfrac{1}{c^2}$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Given: Right $\triangle ABC$ with $\angle ABC = 90°$ and $BD \perp AC$ ($\angle ADB = \angle BDC = 90°$).
We know the AA Similarity Criterion and Pythagoras Theorem:

Formula / Theorem Used: A perpendicular from the right-angle vertex to the hypotenuse forms two triangles similar to the whole triangle and to each other.

Step 1: Proving $\triangle ADB \sim \triangle ABC \sim \triangle BDC$ [2 Marks]
1. In $\triangle ADB$ and $\triangle ABC$: $\angle A = \angle A$ (common) and $\angle ADB = \angle ABC = 90° \implies \triangle ADB \sim \triangle ABC$ (AA).
2. In $\triangle BDC$ and $\triangle ABC$: $\angle C = \angle C$ (common) and $\angle BDC = \angle ABC = 90° \implies \triangle BDC \sim \triangle ABC$ (AA).
Hence, $\triangle ADB \sim \triangle ABC \sim \triangle BDC$ (and consequently $\triangle ADB \sim \triangle BDC$).

Step 2: Deducing (i) $BD^2 = AD \cdot CD$ and (ii) $AB^2 = AD \cdot AC$ [2 Marks]
(i) Since $\triangle ADB \sim \triangle BDC$: $\dfrac{AD}{BD} = \dfrac{BD}{CD} \implies \boxed{BD^2 = AD \cdot CD}$.
(ii) Since $\triangle ADB \sim \triangle ABC$: $\dfrac{AD}{AB} = \dfrac{AB}{AC} \implies \boxed{AB^2 = AD \cdot AC}$.

Step 3: Proving (iii) $\dfrac{1}{p^2} = \dfrac{1}{a^2} + \dfrac{1}{c^2}$ [1 Mark]
From $\triangle ADB \sim \triangle ABC$: $\dfrac{BD}{BC} = \dfrac{AB}{AC} \implies \dfrac{p}{a} = \dfrac{c}{AC} \implies AC = \dfrac{ac}{p}$.
By Pythagoras Theorem (or adding $AB^2 = AD \cdot AC$ and $BC^2 = CD \cdot AC$), $AC^2 = a^2 + c^2$. Substituting $AC = \dfrac{ac}{p}$:

$$\left(\frac{ac}{p}\right)^2 = a^2 + c^2 \implies \frac{1}{p^2} = \frac{a^2 + c^2}{a^2 c^2}$$

$$\boxed{\therefore \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{c^2} \quad \text{[Hence Proved]}}$$

Q13. Case Study 08: Drone Aerial Surveying & Ground Distance Resolution

A surveying drone hovers at a fixed altitude of 120 m directly above a reference landmark. Its wide-angle downward camera captures a high-resolution image where two survey pins on level ground, P and Q, appear separated by a sensor focal distance of 6 cm. The camera's focal length is 50 mm (5 cm). [4]

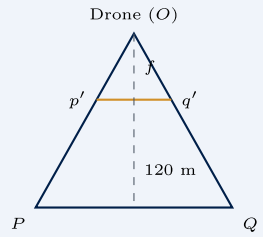
(i) **[1 Mark]** State the linear scale factor between the camera sensor image and the ground distances.

(ii) **[1 Mark]** What is the actual horizontal ground distance between survey pins P and Q?

(iii) **[2 Marks]** If another pair of ground markers is known to be 180 m apart, what distance will separate their images on the camera sensor?

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given Data: Drone altitude $H = 120$ m $= 12000$ cm; camera focal length $f = 5$ cm; sensor separation $p'q' = 6$ cm.
Since the sensor plane is parallel to level ground, by AA similarity ($\triangle Op'q' \sim \triangle OPQ$):



Formula / Theorem Used: Linear Scale Factor $k = \dfrac{\text{Sensor Distance}}{\text{Ground Distance}} = \dfrac{f}{H}$

Sub-part (i): Linear Scale Factor Between Sensor Image and Ground [1 Mark]

Converting both distances to centimetres (1 m = 100 cm $\implies H = 120 \times 100 = 12000$ cm):

$$k = \frac{f}{H} = \frac{5 \text{ cm}}{12000 \text{ cm}} = \frac{1}{2400} \implies \boxed{\text{Scale Factor } = \mathbf{1 : 2400} \left(\frac{1}{2400}\right)}$$

Sub-part (ii): Actual Horizontal Ground Distance PQ [1 Mark]

Using the linear scale factor $k = \dfrac{1}{2400}$ and sensor separation $p'q' = 6$ cm:

$$\frac{6 \text{ cm}}{PQ} = \frac{1}{2400} \implies PQ = 6 \times 2400 \text{ cm} = 14400 \text{ cm} = \frac{14400}{100} \text{ m} = \boxed{\mathbf{144} \text{ m}}$$

Sub-part (iii): Sensor Separation for Ground Markers 180 m Apart [2 Marks]

Given ground distance = 180 m = 18000 cm. Let sensor separation be d cm.
Applying $\dfrac{d}{\text{Ground Distance}} = \dfrac{1}{2400}$:

$$d = \frac{18000}{2400} \text{ cm} = \frac{180}{24} \text{ cm} = \frac{15}{2} \text{ cm} = 7.5 \text{ cm} \implies \boxed{d = \mathbf{7.5} \text{ cm } (75 \text{ mm})}$$

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